

Facial Demorphing from a Single Morph Using a Latent Conditional GAN

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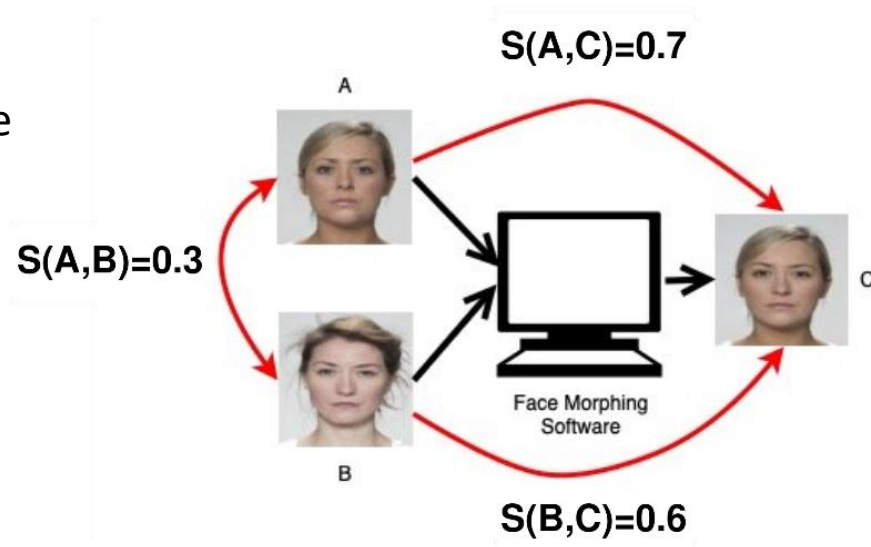
Face Morphing

What is a Face Morph?

“Face morphing is an image manipulation technique where two or more subjects’ faces are blended together to form a single face in a photograph.”

“Morphs can be used to fool both humans and current face recognition systems.”

- from NISTIR 8430



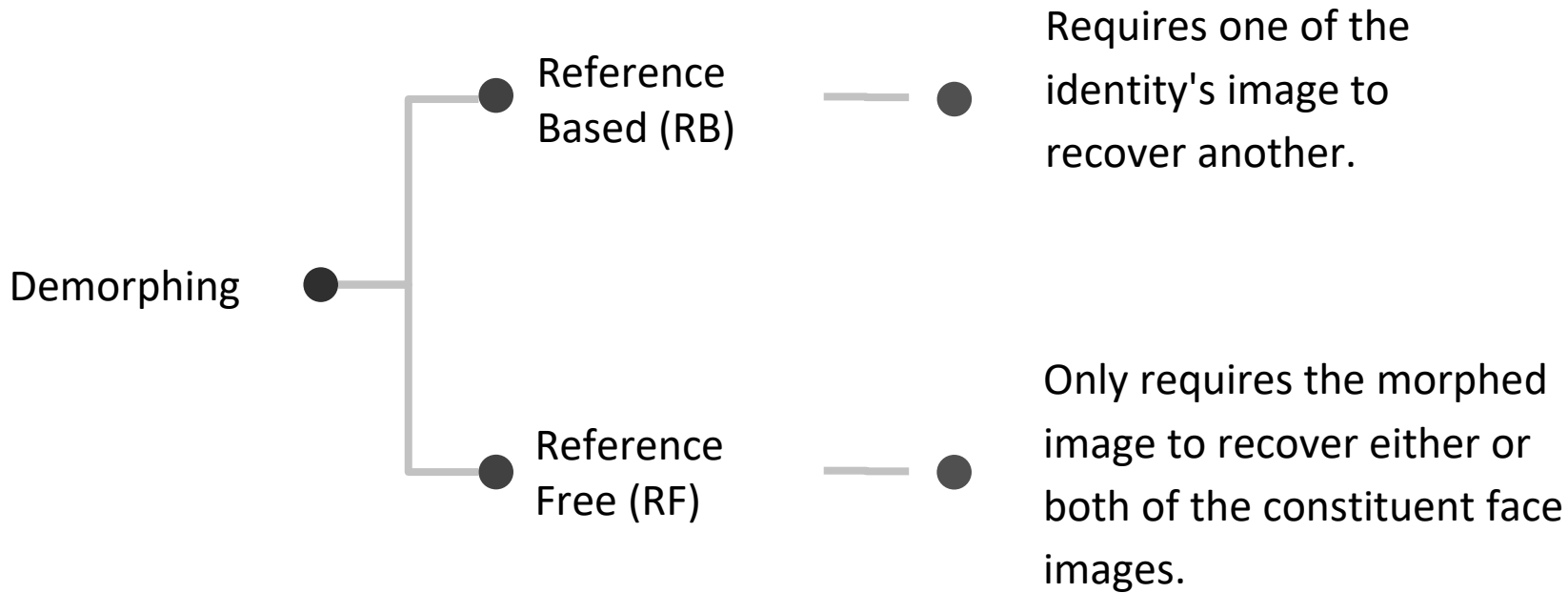
Morph Attacks

Morphed face images can be used covertly in identification documents to allow **multiple individuals to gain access** using a single document.



Face Demorphing

Inverts the process of Morphing!



Need to **recover images** used to create the morph along with the identity.

Challenges in Existing Work (RF)

In this work, we mainly focus on Reference Free Demorphing methods.

Year	Method	Limitation
2022	Face Demorphing [1]	morph-replication
2023	SDeMorph [52]	Train and test morphs share same identities
2024	Identity Preserving Demorphing [54]	Train and test morphs share same identities
2025	dc-GAN [4]	Train and test morphs use the same morphing technique

[1] Banerjee et al., Facial de-morphing: Extracting component faces from a single morph. IJCB 2022.

[52] Nitish Shukla, Sdemorph: Towards better facial de-morphing from single morph. IJCB 2023.

[54] Shukla et al., Facial demorphing via identity preserving image decomposition. IJCB 2024.

[4] Shukla et al. dc-GAN: Dual-Conditioned GAN for Face Demorphing From a Single Morph. FG 2025.

Key Observation

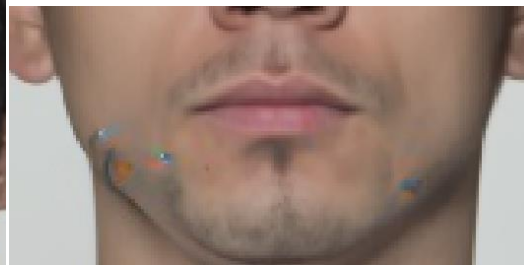
- Previous GAN-based approaches [1,4] directly demorph the morph image.
- Since **each morphing technique introduces distinct artefacts**, separate models are trained per dataset.
- We propose removing irrelevant information in RGB space prior to demorphing.



OpenCV



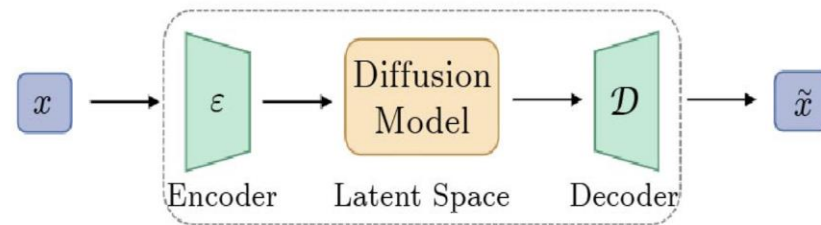
MorDiff



- [1] Banerjee et al., Facial de-morphing: Extracting component faces from a single morph. IJCB 2022.
- [4] Shukla et al. dc-GAN: Dual-Conditioned GAN for Face Demorphing From a Single Morph. FG 2025.

Overview: LDM

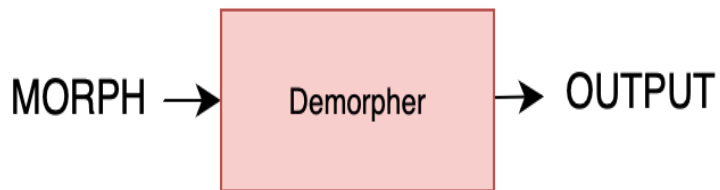
Inspired by Latent Diffusion Models (LDM) [9], we perform demorphing in the latent space, enabling artifact suppression and improved reconstruction.



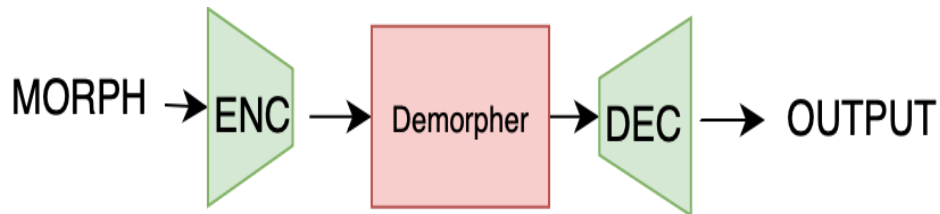
Demorphing is done in a democratized latent space.
Only one demorpher is learned!

Key Idea

In contrast to previous methods, where demorphing is done in pixel domain, **we perform demorphing in latent space guided by the morph.**

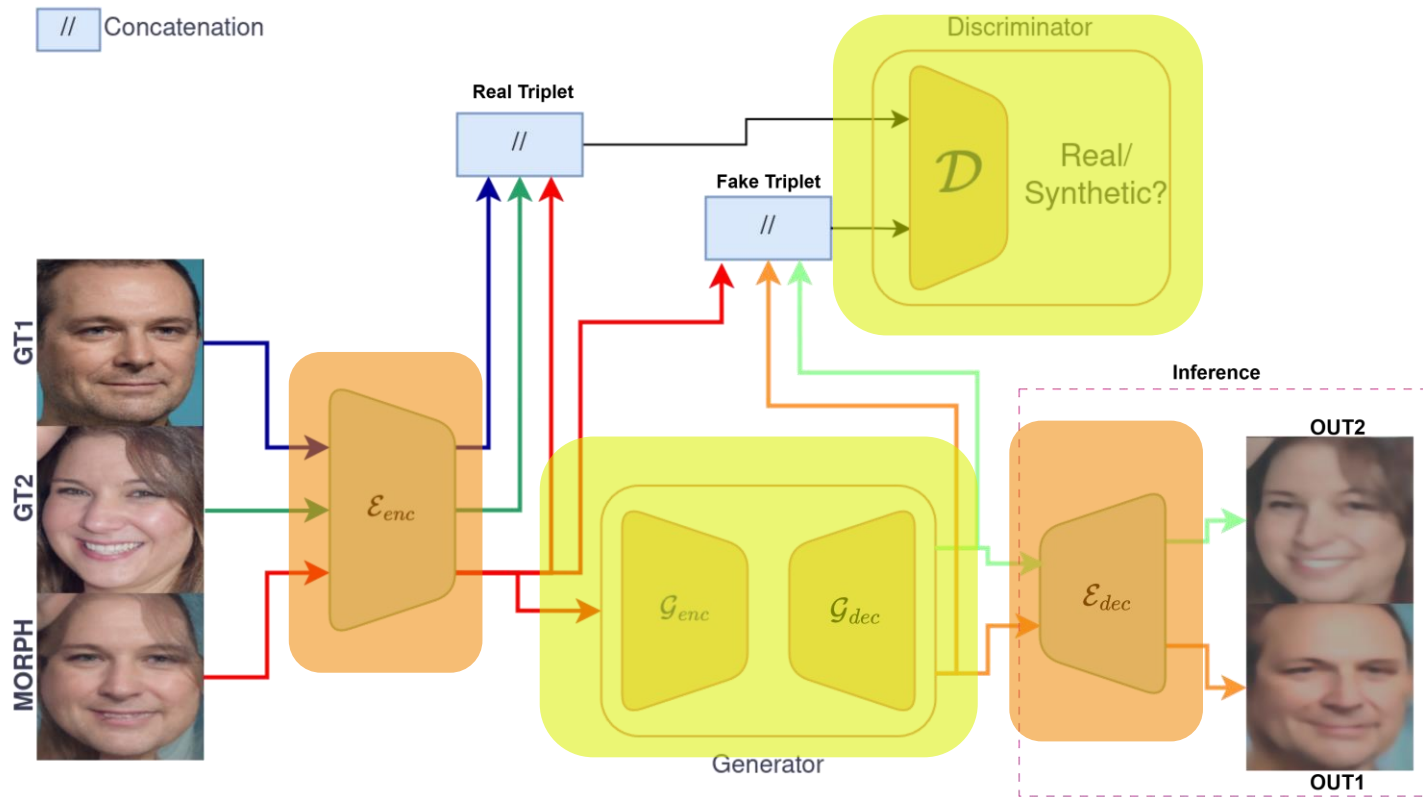


Previous Paradigm

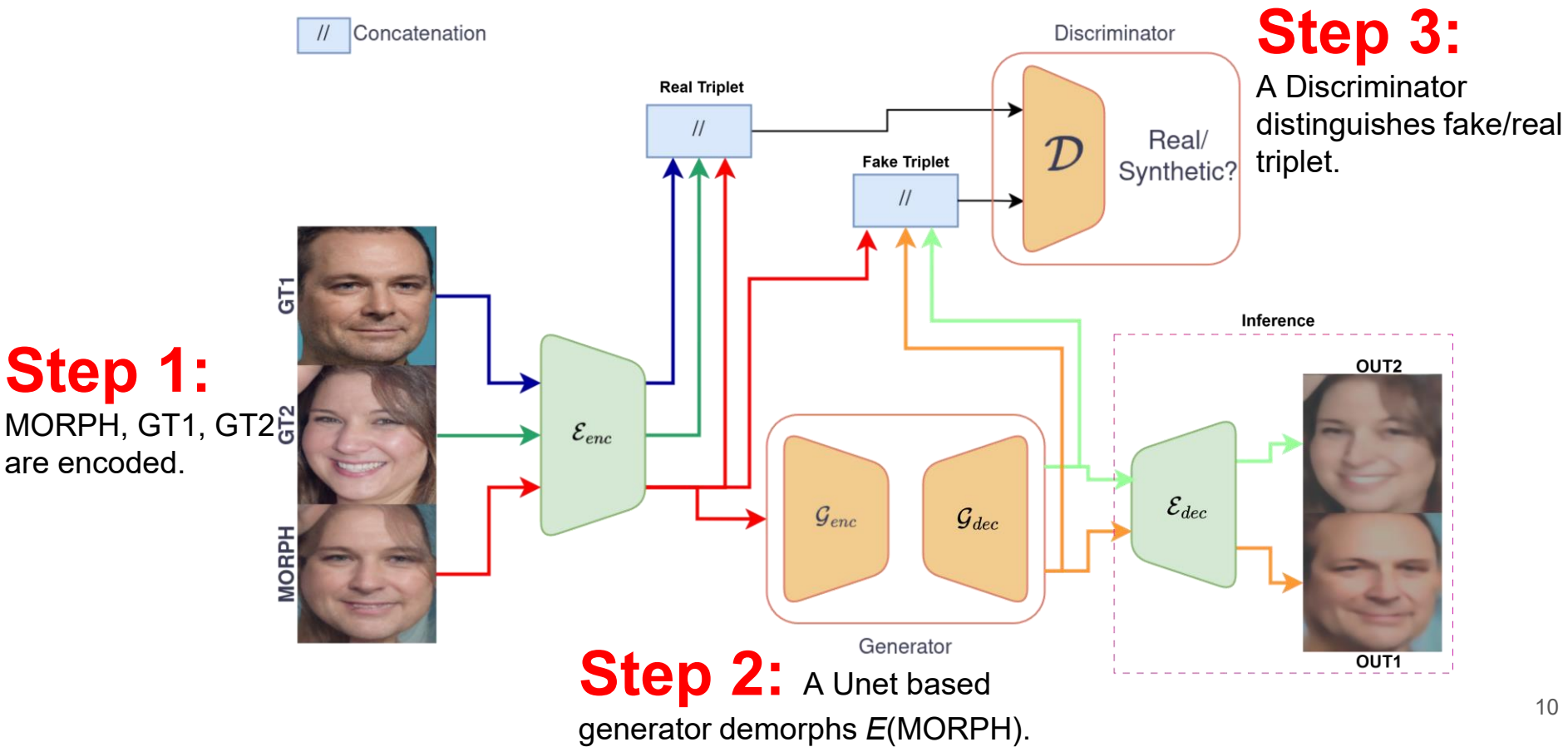


Proposed Paradigm

Latent Conditional GAN : Training



Latent Conditional GAN : Training



Training process

A UNet with 4 input and 8 output channels is trained to demorph the encoded morph Image.

$$\ell = \ell_{cGAN} + \lambda_1 \ell_1 + \lambda_2 \ell_{kurt}$$

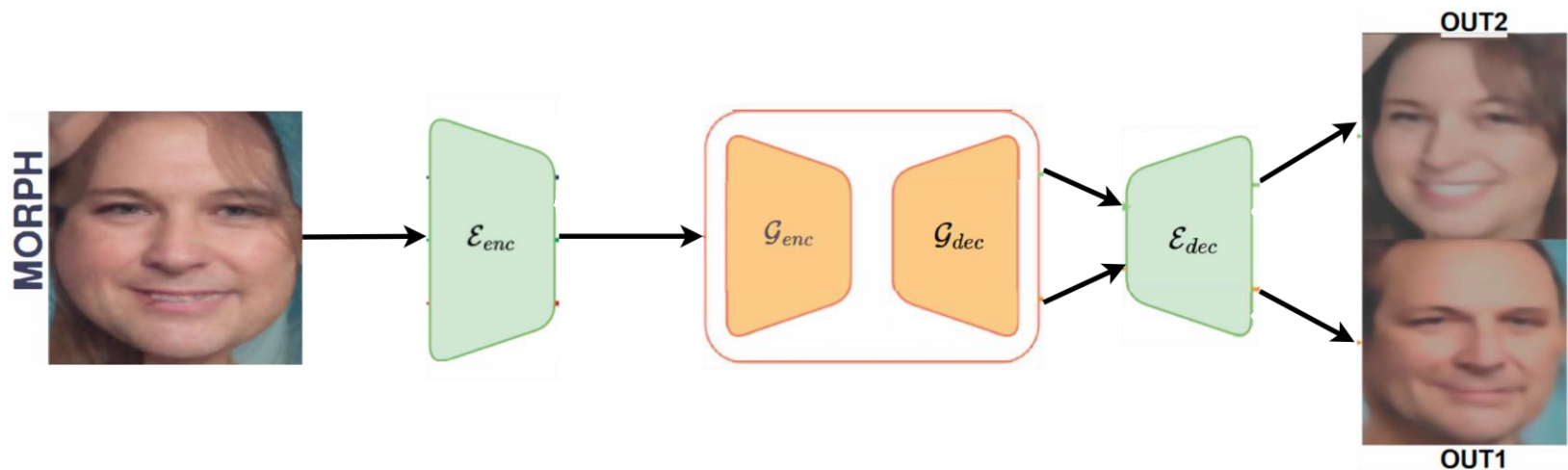
conditional GAN loss

Kurtosis regularization

Encoder: Autoencoder KL [9]
Input (x) dimensions: 3x512x512
Encoded output dimension: 4x64x64

Optimizer: **Adam**
Learning rate: 10^{-4}
Epochs: **300**

Inference process



A face morph is encoded, split into two latent codes by a generator, and decoded to reconstruct the two source faces as outputs.

Dataset Statistics

Training (synthetic)

- Parent Dataset: SMDD Bonafide Set (**synthetic**)
- Morph Generation:
 - Pair of face images
→ morphed using OpenCV/dlib
 - Landmark detection via Dlib
- Preprocessing:
 - Face detection: MTCNN
- Crop, normalize, resize to 256×256
 - Discard images where face not detected
- Total Morphs: **15,000**
- No spatial transformations: preserve feature alignment (lips, nose)

Testing (real)

Dataset	Morph Type	No. of Morphs
AMSL	Landmark-based	2,175
OpenCV	Landmark-based	1,221
StyleGAN	DL-based	1,222
WebMorph	Landmark-based	1,221
FaceMorph	Landmark-based	1,222
MorDiff	DL-based	1,000

We use two face matchers for evaluation:

1. ArcFace [11]
2. AdaFace [20]

[11] Deng, et al., Arcface: Additive angular margin loss for deep face recognition. CVPR 2019.

[20] Kim et al., Adaface: Quality adaptive margin for face recognition. CVPR 2022.

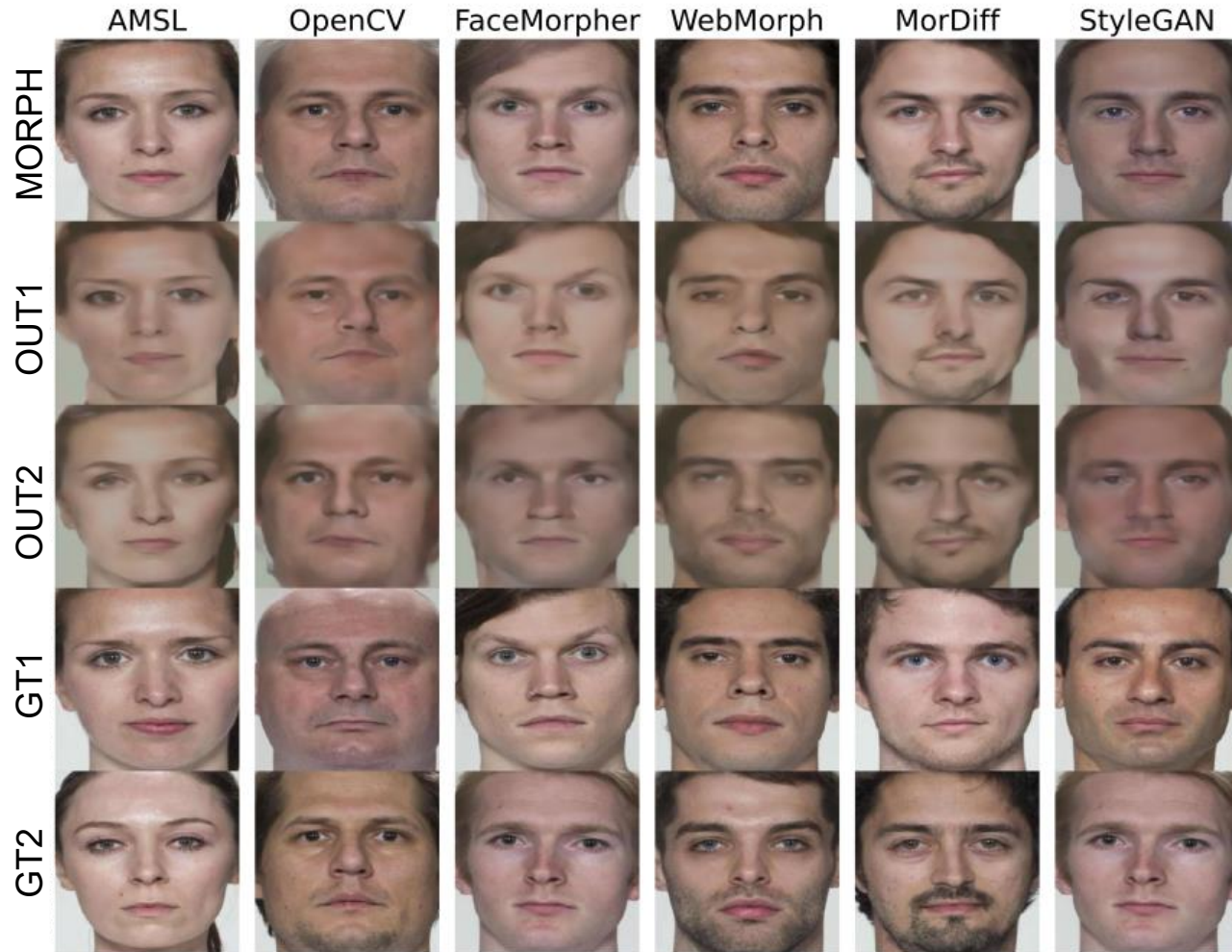
Evaluation Metrics

We evaluate our method on these metrics:

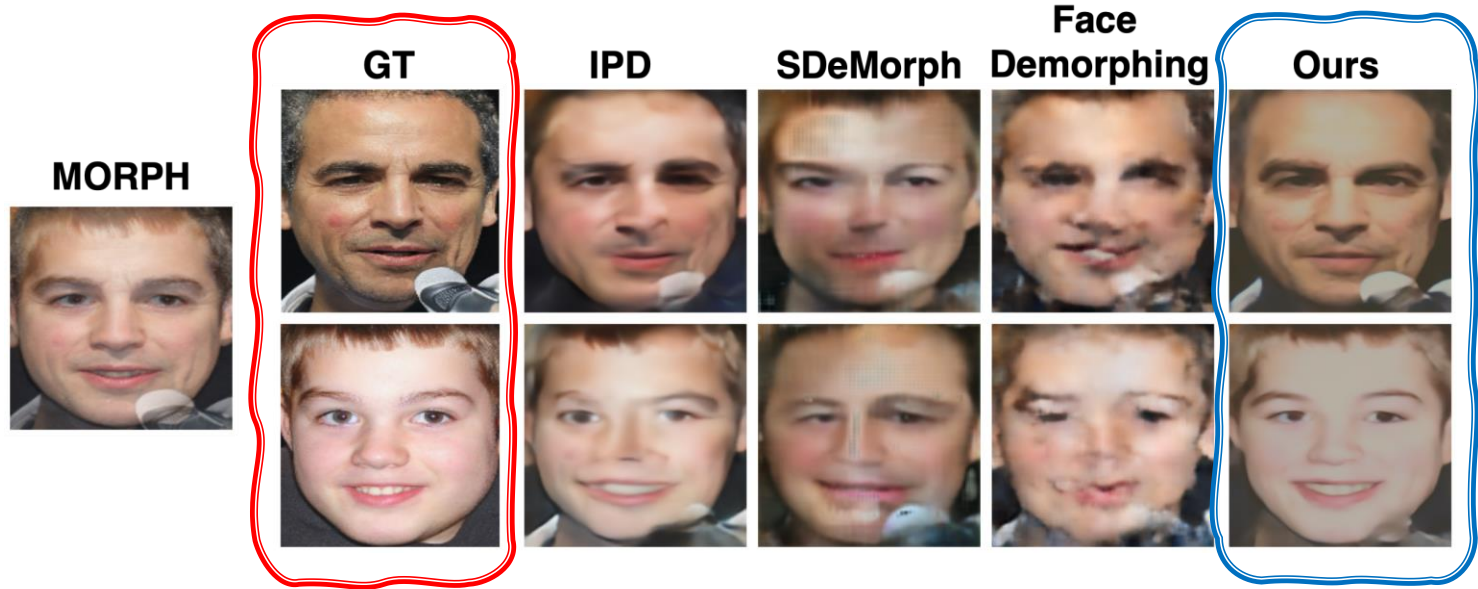
- Restoration Accuracy: fraction of outputs matching ground-truth images based on fixed threshold.
- Image Quality Metrics: SSIM (Structural Similarity Index Measure) and PSNR (Peak Signal-to-Noise Ratio)
- Biometrically Weighted Image Quality Assessment (BW-IQA) [55]

Qualitative Results

1. Demorphed outputs match the ground truth.
2. Demorphed outputs do not exhibit morph replication.

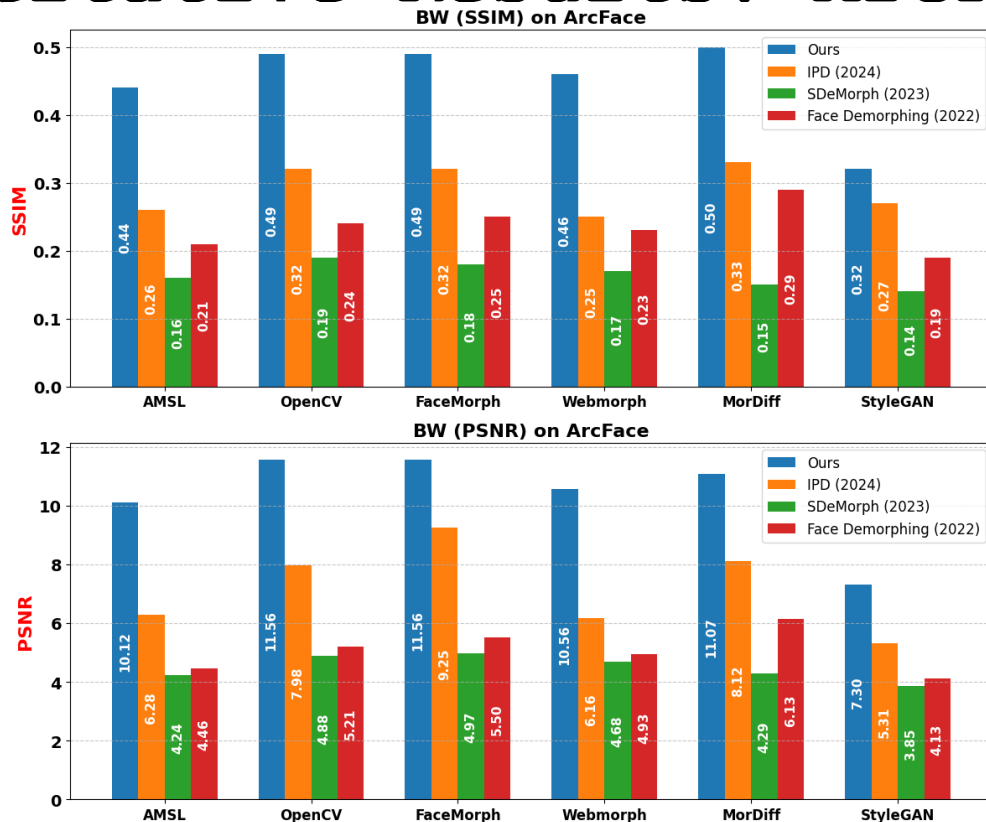


Qualitative Results



Visual comparison with current state-of-the-art:
Our method surpasses current methods in terms of visual quality!

Quantitative Results: ArcFace



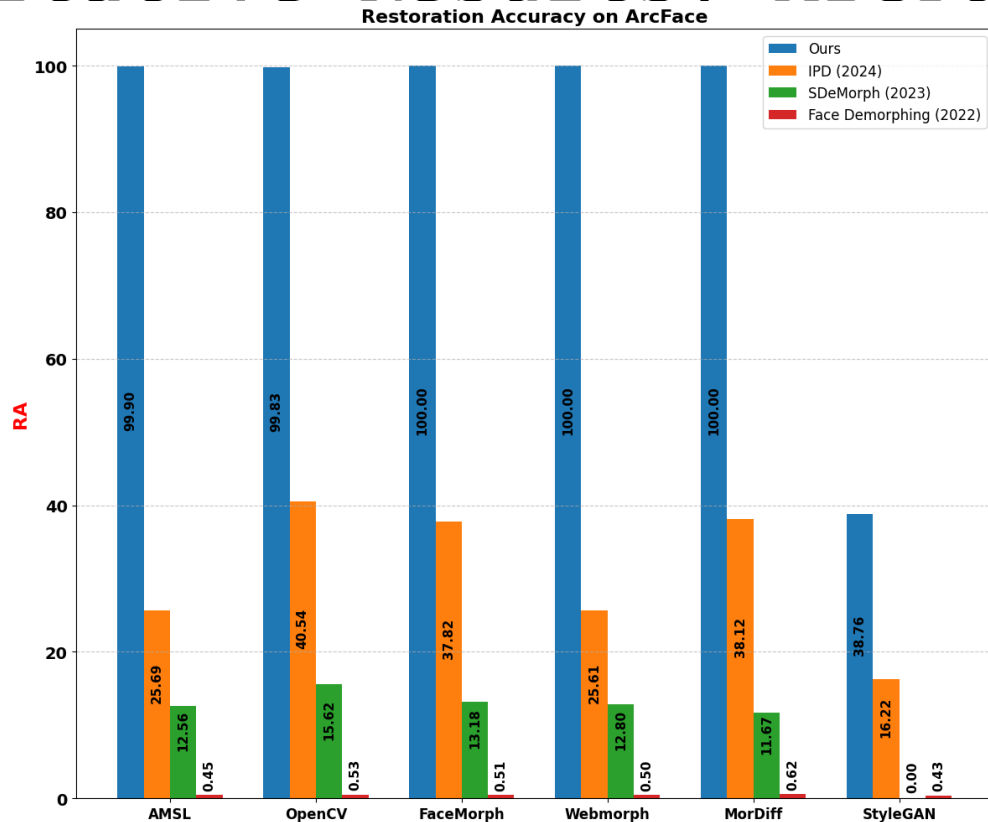
Our methods achieves best biometrically-weighted IQA scores on ArcFace.

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Quantitative Results: ArcFace



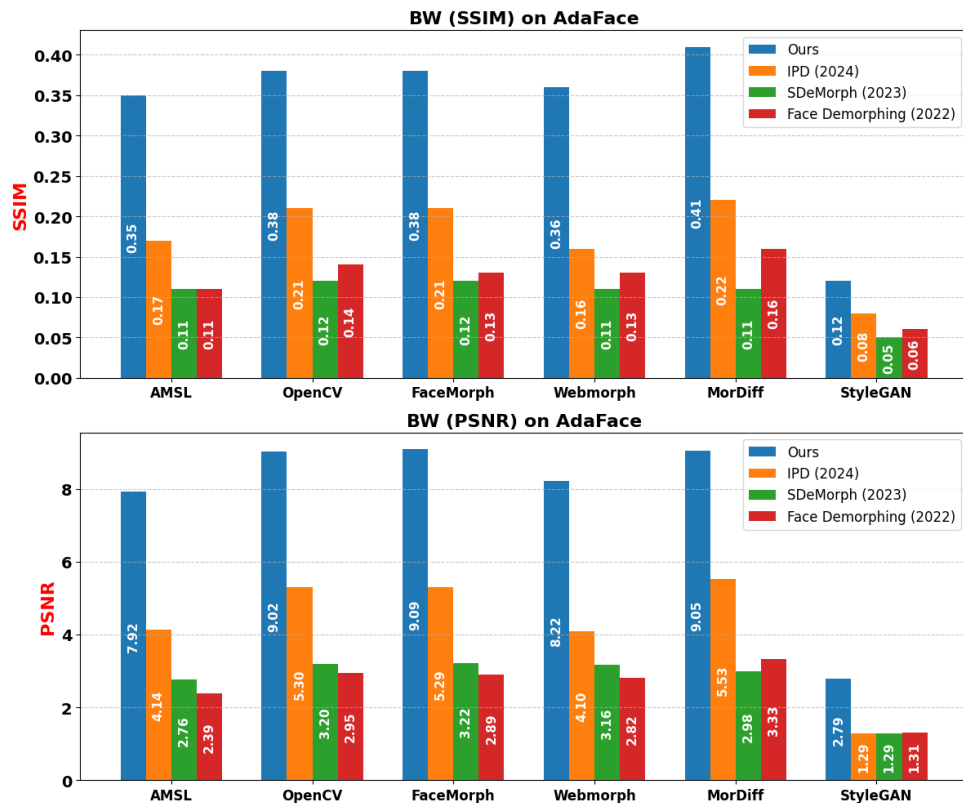
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Quantitative Results: AdaFace



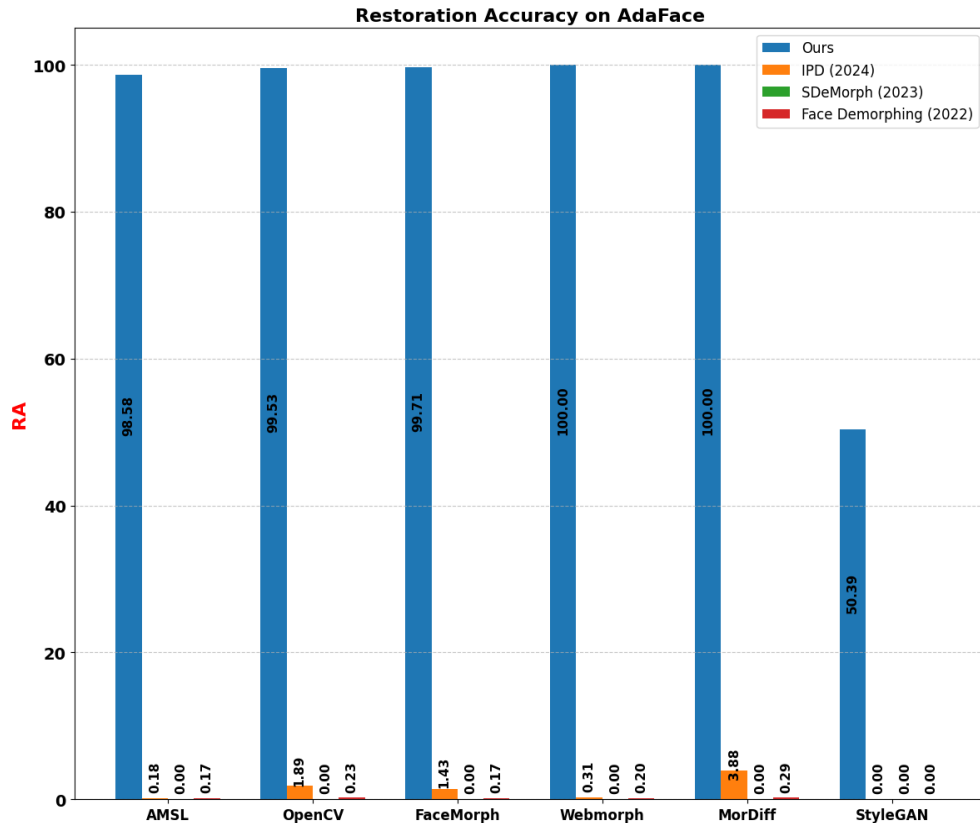
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Quantitative Results: AdaFace



Our methods achieves
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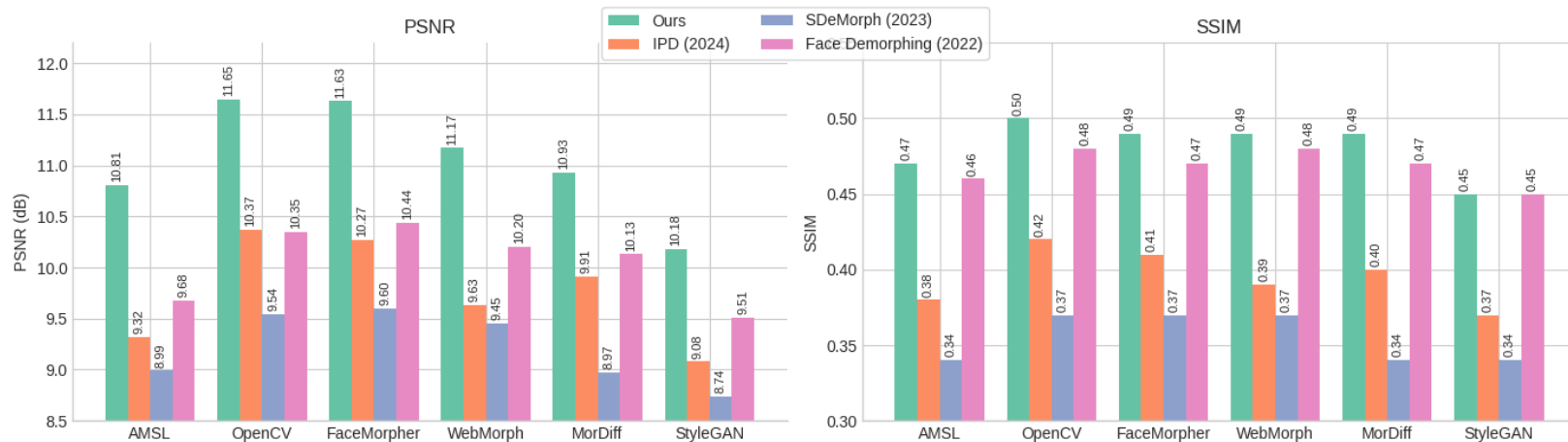
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Quantitative Results

RA @ FMR	AdaFace						ArcFace					
	AMSL	OpenCV	FMorph	Wmorph	MorDiff	StyleGAN	AMSL	OpenCV	FMorph	Wmorph	MorDiff	StyleGAN
10	99.90	99.83	100	100	100	38.76	98.58	99.53	99.71	100	100	50.39
1	99.09	99.30	99.03	99.47	100	12.57	96.35	98.58	98.57	98.91	100	20.75
0.1	96.26	95.83	95.16	96.28	98.63	2.12	91.19	93.69	93.41	94.53	98.83	4.52

Restoration Accuracy with distinct threshold corresponding to 0.1, 1 and 10% FMR.

Quantitative Results



Method	AMSL	OpenCV	FaceMorpher	WebMorph	MorDiff	StyleGAN
Ours	10.81 / 0.47	11.65 / 0.50	11.63 / 0.49	11.17 / 0.49	10.93 / 0.49	10.18 / 0.45
IPD (2024) [54]	9.32 / 0.38	10.37 / 0.42	10.27 / 0.41	9.63 / 0.39	9.91 / 0.40	9.08 / 0.37
SDeMorph (2023) [52]	8.99 / 0.34	9.54 / 0.37	9.60 / 0.37	9.45 / 0.37	8.97 / 0.34	8.74 / 0.34
Face Demorphing (2022) [7]	9.68 / 0.46	10.35 / 0.48	10.44 / 0.47	10.20 / 0.48	10.13 / 0.47	9.51 / 0.45

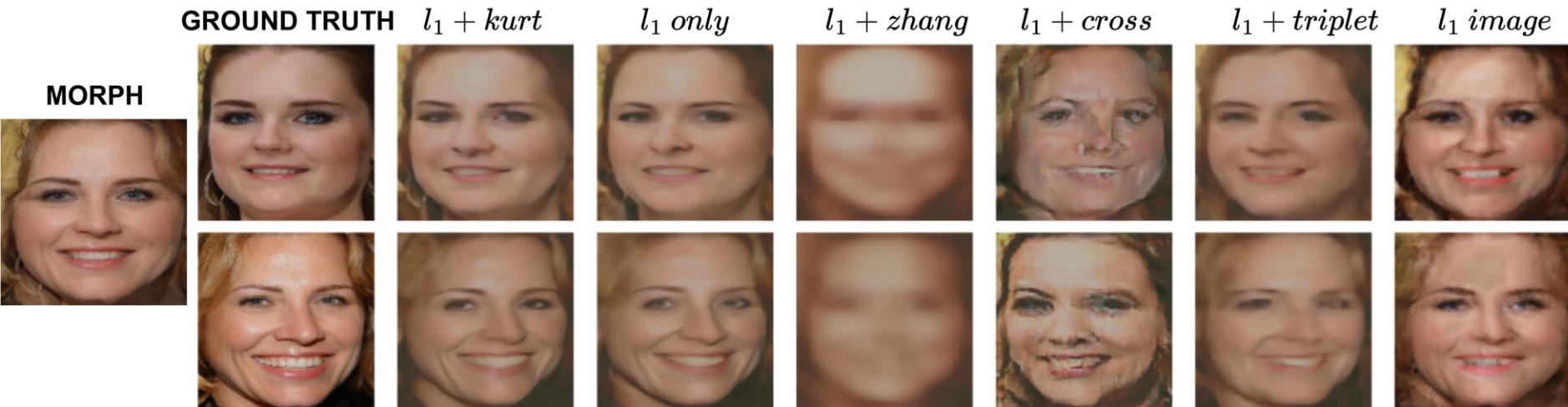
Our method reconstructs high-fidelity images, as reflected by SSIM and PSNR scores.

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Ablation: Separation Priors



Visual comparison of demorphing based on
different image separation priors.

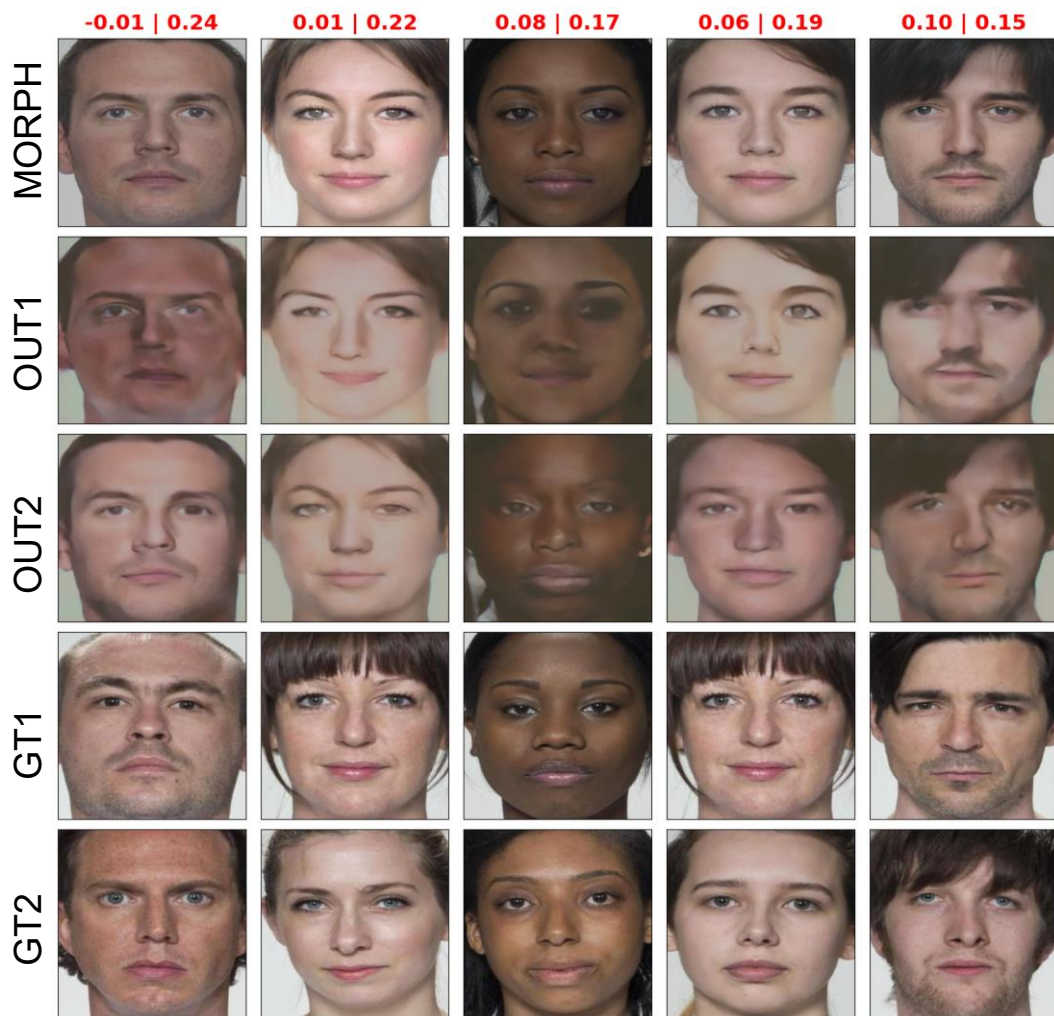
Ablation: Separation Priors

Metric	Dataset	$\ell_1 + \text{kurt}$	ℓ_1 only	$\ell_1 + \text{zhang}$	$\ell_1 + \text{cross}$	$\ell_1 + \text{triplet}$	ℓ_1 image
BW(PSNR) AdaFace/ArcFace	AMSL	7.92/10.12	7.62/10.16	3.30/4.82	2.69/6.80	7.34/9.86	6.86/8.40
	OpenCV	9.02/11.56	8.55/10.98	3.41/5.14	2.84/7.07	7.99/10.23	7.53/9.26
	FaceMorpher	9.09/11.56	8.72/10.86	3.43/5.38	2.85/7.26	7.89/10.43	7.30/9.38
	WebMorph	8.22/10.56	7.88/10.36	3.71/5.08	2.86/7.07	7.55/9.81	6.80/8.42
	MorDiff	9.05/11.07	8.66/10.80	4.6/6.4	3.07/6.82	8.57/10.37	6.82/8.58
	StyleGAN	2.79/7.3	2.66/6.99	1.3/3.8	1.36/5.62	2.77/7.09	2.29/5.83
BW(SSIM) AdaFace/ArcFace	AMSL	0.35/0.44	0.33/0.44	0.15/0.21	0.11/0.27	0.32/0.43	0.32/0.40
	OpenCV	0.38/0.49	0.36/0.47	0.15/0.22	0.11/0.27	0.34/0.44	0.34/0.42
	FaceMorpher	0.38/0.49	0.37/0.46	0.15/0.23	0.11/0.27	0.33/0.44	0.33/0.42
	WebMorph	0.36/0.46	0.34/0.45	0.16/0.22	0.11/0.28	0.33/0.43	0.31/0.39
	MorDiff	0.41/0.5	0.39/0.48	0.20/0.28	0.12/0.27	0.38/0.46	0.32/0.40
	StyleGAN	0.12/0.32	0.12/0.30	0.06/0.17	0.05/0.22	0.12/0.31	0.11/0.27

Kurtosis prior forces the outputs to be more discriminative!

Failure Cases

AdaFace/ArcFace similarity scores between demorphed outputs and ground-truth images indicate room for improvement.



Summary

- Introduces a reference-free **Latent Conditional GAN** that demorphs a single morph by operating and training in a perceptually equivalent latent space
- **Agnostic to morphing technique**, enabling generalization to unseen morph styles without retraining, and scalable to high-resolution images via latent-space compression with a frozen encoder
- **Mitigates morph replication** and suppresses artifacts, yielding higher-fidelity recovered faces and significantly outperforming prior demorphing methods across standard metrics and datasets
- **Trained on synthetic-face morphs and tested on real-face morphs** across techniques
- Future work will target morphs composed of more than two identities

Thank You!

